

The logo of Gazi University is a circular seal. It features the university's name in Turkish, "GAZİ ÜNİVERSİTESİ", around the top inner edge and the year "1926" at the bottom. In the center is a stylized signature of "Gazi".

GAZİ UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING

Lecturer : Dr Ercan Ezin

IE104-COMPUTER PROGRAMMING I

WEEK 14: EXERCISE ATTENDANCE PROGRAM

Exercise



We will implement an attendance displayer program

Exercise: Attendance Record Analyzer

Develop a **C# Console Application** that reads attendance information from a text file named:

```
attendance.txt
```

Each line in the file contains:

```
Date,Week,StudentId
2026-02-16,Week 2,24112022031
2026-02-23,Week 3,24112022031
```

Program Requirements

Your program must:

- Read attendance records from `attendance.txt`
- Ask the user to enter a student ID
- Find all attendance records belonging to that student
- Calculate:
 - Total attended weeks
 - Total course weeks
 - Attendance percentage
- Display all attendance details

Download attendance.txt from course home page.

Example Output

```
Enter your student ID: 241120220
```

```
===== ATTENDANCE REPORT =====
```

```
Student ID: 241120220.
Attended Weeks: 3
Total Course Weeks: 10
Attendance Percentage: 30.00%
```

```
Attendance Details:
```

```
Week 2 - 2026-02-16
```

```
Week 3 - 2026-02-23
```

```
Week 4 - 2026-03-02
```

Technical Requirements

Your solution must include:

- Variables and data types
- Conditional statements
- Loops
- Methods / Functions
- File reading operations

EXAMPLE STRUCTURE

```
using System;
using System.Collections.Generic;
using System.IO;

namespace AttendanceAnalyzer
{
    class Program
    {
        static void Main(string[] args)
        { ... }

        static string GetStudentIdFromUser()
        { ... }

        static void AnalyzeAttendance(string filePath, string studentId)
        { ... }

        static void ShowAttendanceReport(string studentId, int totalWeeks, List<string> studentAttendance)
        { ... }
    }
}
```

```
static void Main(string[] args)
{
    string filePath = "attendance.txt";

    if (!File.Exists(filePath))
    {
        Console.WriteLine("Attendance file could not be found.");
        return;
    }

    string studentId = GetStudentIdFromUser();

    try
    {
        AnalyzeAttendance(filePath, studentId);
    }
    catch (Exception ex)
    {
        Console.WriteLine("An error occurred: " + ex.Message);
    }

    Console.WriteLine();
    Console.WriteLine("Press any key to exit...");
    Console.ReadKey();
}
```

```
static string GetStudentIdFromUser()
{
    string studentId = "";

    while (studentId.Length != 11)
    {
        Console.Write("Enter your student ID: ");
        studentId = Console.ReadLine();

        if (studentId.Length != 11)
        {
            Console.WriteLine("Student ID must contain 11 characters.");
        }
    }

    return studentId;
}
```

```
static void AnalyzeAttendance(string filePath, string studentId)
{
    string[] lines = File.ReadAllLines(filePath);

    List<string> uniqueWeeks = new List<string>();
    List<string> studentAttendance = new List<string>();

    for (int i = 1; i < lines.Length; i++)
    {
        string[] parts = lines[i].Split(',');

        if (parts.Length == 3)
        {
            string date = parts[0];
            string week = parts[1];
            string currentStudentId = parts[2];

            if (!uniqueWeeks.Contains(week))
            {
                uniqueWeeks.Add(week);
            }

            if (currentStudentId == studentId)
            {
                studentAttendance.Add(week + " - " + date);
            }
        }
    }

    ShowAttendanceReport(studentId, uniqueWeeks.Count, studentAttendance);
}
```

```
static void ShowAttendanceReport(string studentId, int totalWeeks, List<string> studentAttendance)
{
    int attendedWeeks = studentAttendance.Count;

    if (attendedWeeks == 0)
    {
        Console.WriteLine("No attendance record found.");
        return;
    }

    double attendancePercentage =
        (double)attendedWeeks / totalWeeks * 100;

    Console.WriteLine();
    Console.WriteLine("==== ATTENDANCE REPORT =====");
    Console.WriteLine("Student ID: " + studentId);
    Console.WriteLine("Attended Weeks: " + attendedWeeks);
    Console.WriteLine("Total Course Weeks: " + totalWeeks);
    Console.WriteLine("Attendance Percentage: " +
        attendancePercentage.ToString("0.00") + "%");

    Console.WriteLine();
    Console.WriteLine("Attendance Details:");

    foreach (string attendance in studentAttendance)
    {
        Console.WriteLine(attendance);
    }
}
```




HOMEWORK

Do the same exercise by only using arrays instead of List class.

THE END



CHECK OUT COURSE WEBSITE: [HTTPS://GAZI-END-MUH.GITHUB.IO](https://GAZI-END-MUH.GITHUB.IO)

GOT ANY QUESTIONS LATER?

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